

Lesson Exemplar in General Science

Quarter 3

LE

5

Lesson Exemplar for General Science
Quarter 3: Unit 5

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LESSON EXEMPLAR

Learning Area	GENERAL SCIENCE	Grade Level	11
Semester	SECOND	Quarter	3

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	<i>The learners learn that organ systems work together to carry out essential functions of an organism</i>
Performance Standard	<i>The learners use examples to describe the relationships among living systems and their components, as well as how the structure of a part relates to its function. They explain the role of cells in the perpetuation of life and how organ systems work together to maintain good health. They conduct investigations on the effects of climate change in local ecosystem.</i>
Learning Competencies	Key Topics Learned in Previous Grade Level • describe in simple terms how the following human body systems work: muscular, skeletal, digestive, circulatory, and respiratory; • identify from pictures and diagrams the parts of the circulatory system as heart, blood, and blood vessels, and describe how they work; • use models, flow charts, diagrams, and simulations to explain how body systems work together, such as digestion and excretion In this lesson, Grade 11 learners will be able to: • Describe how damaged part in a system affects the organism using the case of torn ACL (skeletal and muscular) and asthma attack (respiratory and circulatory)
II. REFERENCES and MATERIALS <i>(Selecting Resources)</i>	• Amoeba Sisters. “Human Body Systems Functions Overview: The 11 Systems.” YouTube video, 6:52. Posted May 28, 2017. https://www.youtube.com/watch?v=gEUu-A2wfSE • Peekaboo Kidz. <i>How Your Organs Work Together to Keep You Alive: A Deep Dive</i>. YouTube video, 5:46. Posted October 14, 2020. https://www.youtube.com/watch?v=c4ZxPG_WxOA

<i>and Material)</i>	- Learn Easy Science. 12 Organ Systems Roles & Functions Easy Science Lesson. YouTube video, 5:58. Posted May 7, 2021. https://www.youtube.com/watch?v=cQIU0yJ8RBg - Ochi, M., K. Shino, K. Yasuda, and M. Kurosaka, eds. <i>ACL Injury and Its Treatment</i>. Tokyo: Springer, 2016. https://doi.org/10.1007/978-4-431-55858-3 - Levitzky, Michael G. <i>Pulmonary Physiology</i>. 8th ed. New York: McGraw-Hill Education, 2013.	
III. CONTENT <i>(Sequencing Content)</i>	<i>Organ systems working together</i> <i>Overview of human body system</i> <i>Skeletal and muscular system interaction Key</i> <i>Respiratory and circulatory system interaction organs and tissues in each system</i> <i>Interconnectedness of Human Body Systems and the Impact of Localized Damage on Whole-Body Function</i>	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	<i>Unpacked Learning Competencies</i> <i>Identify the basic functions of the skeletal, muscular, respiratory, and circulatory systems</i> <i>Explain how two body systems interact to perform a function (e.g., movement or oxygen delivery)</i> <i>Compare the effects of damage in different systems (e.g., ACL vs. asthma)</i> <i>Illustrate how a specific injury or condition affects system function</i>	
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i>		ANNOTATION <i>*Instruction to the teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge	1. Activating Prior Knowledge <i>To activate learners' prior knowledge, the teacher begins the lesson by guiding students through a brief overview of the human body systems. The teacher highlights the basic functions of the skeletal, muscular, respiratory, and circulatory systems, ensuring that students recall how</i>	Notice to the Teacher <i>* The teacher will choose one activity from three options. After the students complete the assigned activity, they will answer the guide</i>

	each system contributes to movement, oxygen delivery, and overall body function Option 1: Body System Brain Dump The teacher will implement a cooperative learning strategy by dividing students into small groups of 3–4 members. Each group will be given three (3) minutes to list everything they already know about human body systems—including system names, related organs, and their functions—on a sheet of paper. After the time is up, the teacher will facilitate a brief class discussion to review common ideas, clarify misunderstandings, and connect students’ existing knowledge to the upcoming lesson. Note: Expected Correct Responses: <table><tr><th>Body System</th><th>Correctly Sorted Terms</th></tr><tr><td>Skeletal System</td><td>Bones, joints, skull, ribs, vertebrae, supports the body, protects internal organs</td></tr><tr><td>Muscular System</td><td>Muscles, tendons, biceps, contracts to move (voluntarily), enables movement, stabilizes joints</td></tr><tr><td>Respiratory System</td><td>Lungs, trachea, diaphragm, breathes in oxygen, breathes out carbon dioxide, gas exchange</td></tr><tr><td>Circulatory System</td><td>Heart, blood vessels, arteries, veins, pumps blood, carries oxygen, delivers nutrients</td></tr></table> Process Questions: 1. How does this organ or function help the body system work? 2. What would happen to the whole body if one of these systems is damaged? Option 2: Think-Pair-Share:	Body System	Correctly Sorted Terms	Skeletal System	Bones, joints, skull, ribs, vertebrae, supports the body, protects internal organs	Muscular System	Muscles, tendons, biceps, contracts to move (voluntarily), enables movement, stabilizes joints	Respiratory System	Lungs, trachea, diaphragm, breathes in oxygen, breathes out carbon dioxide, gas exchange	Circulatory System	Heart, blood vessels, arteries, veins, pumps blood, carries oxygen, delivers nutrients	questions. The teacher facilitates the activity and provides constructive feedback IDF Element * The IDF elements present in this section are reflective, in nature. It is reflective because the activity focuses on critical thinking, drawing from related experiences. It is in Analysis phase because it involves identifying learning goals, understanding learners' needs, analyzing tasks, and setting clear instructional objectives This is ideational because it emphasizes the things or characteristics that students see and experience (context) in order to perform the activity. The guide questions also serve as a formative assessment, allowing the teacher to evaluate the breadth and depth of students' understanding of the concept This belongs to the engage phase, where learners actively connect with the topic, explore ideas, and build curiosity before diving deeper into the lesson.
Body System	Correctly Sorted Terms											
Skeletal System	Bones, joints, skull, ribs, vertebrae, supports the body, protects internal organs											
Muscular System	Muscles, tendons, biceps, contracts to move (voluntarily), enables movement, stabilizes joints											
Respiratory System	Lungs, trachea, diaphragm, breathes in oxygen, breathes out carbon dioxide, gas exchange											
Circulatory System	Heart, blood vessels, arteries, veins, pumps blood, carries oxygen, delivers nutrients											

	<i>The teacher asks the students to pair up and reflect on everyday actions and to discuss which systems are involved. The teacher encourages the students to share their ideas with the whole class. Together, they list the body systems involved in those actions.</i> <i>Process Questions:</i> <i>1. Which body systems do you think are working together in that action?</i> <i>2. How do these systems depend on one another to keep the body functioning?</i> 2. Establishing the Purpose of the Lesson <i>The teacher begins by revisiting the basic functions of the skeletal, muscular, respiratory, and circulatory systems, as discussed in the previous activity. To deepen students' understanding, the teacher introduces a guiding thought: "The body doesn't rely on one system alone—every part works together to keep us moving, breathing, and alive"</i> Option 1: "One System Stops", Action <i>Students form a small group to act out short scenes showing what might happen if one body system suddenly stops working for 30-60 seconds. Each group is assigned a system—skeletal, muscular, respiratory, or circulatory. The teacher encourages exaggerated but clear movements to highlight the impact.</i> <i>Sample prompts:</i> <i>• "Your muscles stop working—how would you try to move?"</i> <i>• "Your lungs can't take in air—how would you react?"</i> <i>• "Your bones can't support your body—how would you stand or walk?"</i>	<i>* This section is both responsive and reflective as it recognizes the diverse backgrounds of students while emphasizing critical thinking and metacognition. It fosters an inclusive learning environment, encouraging students to analyze concepts deeply and connect their experiences to broader ideas.</i> <i>* The teacher will choose one activity from two options. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>The teacher used the activity as formative assessment because it gathers informal data about students' prior knowledge and misconceptions .and helps tailor the lesson to student needs Also, this shows a metacognitive strategy encouraging the students to evaluate their thought processes.</i>
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	<i>Process Questions</i> 1. <i>What real-life situations might look like this?</i> 2. <i>Why is it important for body systems to work together?</i> Options 2: Quick Poll <i>To begin the lesson, the teacher asks a question, “Imagine one part of your body—like your heart, lungs, or bones—gets injured or stops working. Do you think that could affect the rest of your body? The teacher then asks students to raise their hand if they think yes, and keep their hand down if they think no. Once the student responded the teacher encouraged the student to volunteer to briefly explain their reasoning.</i> Option 3. Answer Garden “How Does One Body Part Affect the Whole?” <i>To begin the lesson, the teacher introduces a simple, thought-provoking question to the class: “Let’s think about how our body works as a system. Imagine one organ or body part—like your heart, lungs, or bones—stops working. What do you think happens to the rest of the body?”</i> <i>The teacher then explains that students will use a digital tool called AnswerGarden to share their ideas anonymously. Each student will access the AnswerGarden link using a phone, tablet, or computer and type in a short response—just a few words or a phrase. The teacher highlighting key ideas from the word cloud, clarifying misconceptions and connecting student ideas to the lesson.</i>	<i>This option used technology integration allowing anonymous, inclusive participation.</i> <i>Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context.</i>
B. Instituting New Knowledge	1. Presenting Examples <i>The teacher briefly recalls the previous lesson, where students explored how body systems—skeletal, muscular, respiratory, and circulatory—work together to keep the organism alive and functioning. To spark curiosity, the teacher poses a guiding question: “If the body</i>	<i>* The teacher will choose one activity from two options given. After the students complete the assigned activity, they will answer the guide</i>

	is like a team, what happens when one player gets injured—can the rest still win the game? Option 1: Visual Inquiry Briefing The teacher facilitates a Visual Inquiry Briefing to guide students in analyzing how Interconnectedness of Human Body Systems and how two body systems interact to perform a function. 1. Amoeba Sisters. (2017, May 28). Human body systems functions overview: The 11 systems. YouTube. https://www.youtube.com/watch?v=gEUu-A2wfSE This video provides a concise and engaging overview of the human body systems, with a segment that emphasizes how they work together 2. Peekaboo Kidz. (2020, October 14). How your organs work together to keep you alive: A deep dive. YouTube. https://www.youtube.com/watch?v=c4ZxPG_WxOA This video is great for showing how body systems “team up” to perform essential functions 3. Learn Easy Science. (2021, May 7). 12 Organ Systems Roles & functions Easy science lesson [Video]. YouTube. https://www.youtube.com/watch?v=cQIU0yJ8RBg This video offers a clear and accessible breakdown of each organ system’s role—perfect for reinforcing foundational knowledge before diving into system interactions and damage effects. Process Questions 1. How does each system contribute to keeping the body alive and functioning? 2. Why is it important for body systems to work together? 3. Can you think of another example where two systems must cooperate to perform a function? Options 2: Interactive Concept Briefing To deepen students’ understanding of system interdependence, the teacher facilitates an Interactive Concept Briefing using a slide presentation and guided process questions	questions. The teacher facilitates the activity and provides constructive feedback. * This remains a formative assessment, designed to provide insight into students’ understanding. It is integrative, as it takes a broad perspective on concepts, considering various aspects. It is also linked to context and connection, focusing on students’ backgrounds and related experiences. Furthermore, it falls under the explore phase, as the activity allows students to navigate and deepen their knowledge. Design phase focuses on sequencing content, selecting appropriate strategies, and making learning meaningful which ensure coherence and engagement
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Table 1. Body Systems and their Connection to Other Systems

Body System	Main Function	Key Organs/ Structures	Connection to Other Systems
Skeletal System	Provides structure, support, and protection; enables movement with muscles	Bones, joints, ligaments (e.g., ACL)	Works with muscular system for movement; protects organs; supports circulatory and respiratory systems
Muscular System	Enables movement; maintains posture; generates heat	Muscles, tendons	Pulls on bones to move; supports breathing (e.g., diaphragm); affects circulation during activity
Respiratory System	Delivers oxygen to the body and removes carbon dioxide	Lungs, trachea, bronchi, diaphragm	Supplies oxygen to circulatory system; diaphragm works with muscular system; oxygen fuels all systems
Circulatory System	Transports oxygen, nutrients, and waste; regulates body temperature	Heart, blood vessels, blood	Delivers oxygen from respiratory system; supports muscles during movement; affected by injuries or low oxygen

Process Questions:

1. How does each system contribute to keeping the body alive and functioning?
2. If one system speeds up (e.g., heart rate increases), what might happen to the others?
3. Which system do you think is most vulnerable to affecting others when damaged? Why?

2. Discussing New Concept

The teacher reminds the class of their previous exploration into how body systems work together to perform essential functions such as movement, breathing, and circulation

Option 1: Case-Based Concept Mapping

Before beginning the activity, the teacher gathers the class and projects an image.

- a) A basketball player collapsing mid-game, clutching their knee. Rather than simply describing a torn anterior cruciate ligament (ACL), the teacher frames a deeper question: How does one injury affect the entire body?*
- b) A young woman standing on a city sidewalk, clutching her chest and gasping for air, her inhaler raised shakily to her mouth as she struggles through an asthma attack.*



a) Torn ACL



b) Asthma attack

The teacher asks the students to work in pairs using a concept map. At the center is either torn ACL or asthma attack and surrounding it are key components of the skeletal, muscular, circulatory, and respiratory systems. Students will trace the ripple effects of the

** Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.*

This approach is inclusive, ideational, and integrative. The processing questions acknowledge various contexts, facilitate conceptual understanding, and provide a holistic perspective. It also emphasizes context and connection, ensuring that learners relate their experiences to the discussion.

Also, this is inquiry-Based Learning because students explore relationships between body systems rather than receiving direct answers.

The teacher facilitates a discussion comparing the two cases, emphasizing how system interdependence is revealed through injury and recovery.

injury—how it disrupts joint function, triggers muscular compensation, and affects overall mobility

Process Question

1. *What changes occur in the muscular system when the ACL is damaged?*
2. *How do muscles compensate for the loss of ligament support?*
3. *How does reduced oxygen availability affect the heart and blood vessels?*
4. *What symptoms indicate that both systems are under stress (asthma attack)?*
5. *What does this case reveal about the interdependence of body systems?*

Table 2. Skeletal and Muscular Systems

Key Aspect	Skeletal System	Muscular System
Structure Involved	ACL (Anterior Cruciate Ligament) connects femur to tibia within the knee joint	Quadriceps, hamstrings, and surrounding stabilizing muscles
Function Before Injury	Provides joint stability, controls tibial movement, supports weight-bearing	Enables movement, stabilizes joint, responds to neural signals for coordination
Effect of ACL Tear	Loss of joint stability, misalignment of bones, increased risk of further damage	Muscle inhibition, reduced strength, altered movement patterns
Compensatory Response	Other ligaments and bones may bear abnormal stress	Muscles attempt to compensate, often leading to overuse or imbalance
Clinical Consequences	Joint degeneration, possible meniscal/cartilage	Muscle atrophy, weakness, impaired

		damage, osteoarthritis risk	proprioception and coordination	
	Rehabilitation Focus	Restore structural integrity via surgery or bracing	Rebuild strength, retrain neuromuscular control, improve dynamic stability	
	System Interdependence	Bones need muscular support to maintain alignment and function	Muscles rely on stable skeletal structure for effective contraction and movement	
Option 2. Cause-and-Effect Chain Building Activity <i>The class is divided into small groups, each assigned one of two real-life cases: a torn ACL or an asthma attack. Each group receives a strip of blank cards or sticky notes and is instructed to build a cause-and-effect chain or a domino chain of effects that visually represents the sequence of events triggered by the injury or attack. The first card identifies the initial event (e.g., “ACL tear during sudden pivot” or “Airways constrict during asthma attack”), and each subsequent card describes a consequence or system response.</i> <i>Possible answer</i> <i>ACL tear → joint instability → muscle compensation → reduced movement → circulatory impact</i> <i>Airway constriction → Reduced airflow to lungs → Decreased oxygen exchange → Lower oxygen levels in blood → Increased heart rate and circulatory strain</i> <i>Process Questions:</i> <i>What are the short-term effects on movement and coordination?</i> <i>What long-term consequences might arise if the injury is not properly treated?</i>				
<i>*This approach is relevant and responsive, incorporating transdisciplinary learning and problem-solving. It transcends language and the formation of ideas, connecting with various fields of knowledge. Additionally, it is solution-oriented, focusing on addressing challenges</i> <i>Furthermore, because it nurtures multiple skills and areas of expertise, it is integrative—encouraging learners to apply diverse knowledge. The process also fosters collaboration and creativity.</i> <i>*Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. It supports varied learning styles.</i> <i>After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i>				

	3. <i>What role does medication (e.g., inhaler) play in restoring system function?</i> 4. <i>How do the respiratory and circulatory systems coordinate to maintain homeostasis?</i> 3. Developing Mastery <i>s stage of the lesson, students are reminded that mastery of the topic involves recognizing not just what each system does, but how they work together—and what happens when that coordination is disrupted. Through the cases of a torn ACL and an asthma attack, students will explore how damage to one part of the body can ripple across systems, affecting the organism as a whole.</i> Option 1: System Under Stress: Expert Panel Simulation <i>nts take on the roles of medical interns presenting to a panel of health professionals. Their task is to explain how a specific injury or condition disrupts body system coordination and affects the organism’s overall function. They will be given a 3–5-minute presentation and answer one follow-up question from the class.</i> Options 2: Body Systems: Compare, Connect, Reflect <i>The teacher asks the students to create a Venn diagram to compare the two real-life health cases, the torn ACL and Asthma. Students will identify what’s unique to each other, what they share, and how they interact. They will write a 5-7 sentence reflection explaining what they learned, how the cases are connected, and why understanding these connections matters.</i> Process Questions: 1. <i>What are the key differences between the torn ACL and asthma attack in terms of the body systems affected? → Prompts identification of primary and secondary systems involved.</i>	<i>This activity uses a metacognitive pedagogy because the student promotes awareness of thinking and learning process.</i> <i>Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. It supports varied learning styles.</i> <i>After the students complete the assigned activity, they will answer the guide questions. The teacher facilitates the activity and provides constructive feedback.</i>
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2. Which effects are unique to each case, and which are shared across both?
3. Why is it important to understand how body systems work together?

Possible Venn diagram

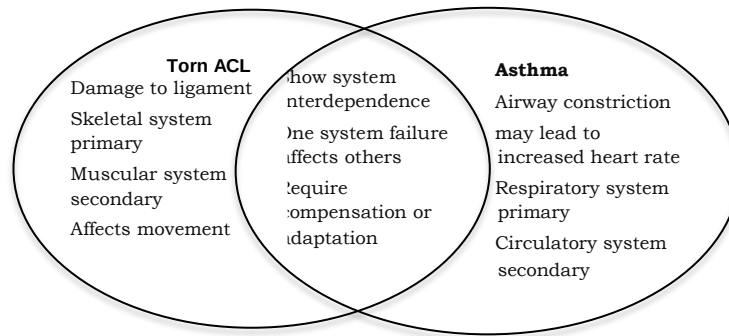


Figure3. Comparing Torn ACL and Asthma using a Venn Diagram

Table 3. Possible Rubric

Criteria	Excellent (5)	Good (3)	Needs improvement (1)
Content Accuracy	All facts are accurate and clearly stated	Most facts are accurate	Many inaccuracies or missing content
Comparison Depth	Strong, detailed comparison with clear overlap	Adequate comparison with some overlap	Minimal comparison or missing sections
Explanation Insight	Thoughtful, well-explained, connects ideas	Explains ideas with some insight	Little to no explanation or insight

C. Demonstrating Knowledge and Skills	1. Finding Practical Application <i>After guiding students through the functions and interactions of body systems, the teacher transitions the class toward real-life relevance. The teacher explains that understanding how systems work together is not just academic—it has practical implications for health, recovery, and decision-making.</i> Option 1. System Snapshot: What’s affected? <i>The teacher introduces a short activity called System Snapshot: What’s Affected? Students will analyze either the torn ACL or asthma attack case and identify which body systems are affected, how they respond, and what the organism experiences as a result. Students are then asked to choose one case and create a simple visual or written snapshot that traces the impact of the damage across systems. A brief explanation of how the systems interact or influence each other</i> <i>Process Questions:</i> <i>1. Which system is trying to compensate for the damage?</i> <i>2. “What symptoms would you expect to see in this case?”</i> Option 2: Which System Needs Help First? <i>Students are given short medical scenarios and asked to decide which system should be prioritized for intervention—and justify their choice.</i> <i>Sample Scenarios</i> <i>1: Torn ACL during a basketball game. The athlete collapses and cannot stand.</i> <i>• Should the focus be on stabilizing the skeletal system or supporting the muscular system first?</i>	<i>*Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. It supports varied learning styles.</i> <i>After the students complete the assigned activity, they will answer the process questions. The teacher facilitates the activity and provides constructive feedback.</i> <i>*This is reflective, as it highlights critical thinking and metacognition in the development of the activity and its explanation. Additionally, the empathize stage is present, where students can express their ideas and connect personally with the subject matter.</i>

	2: Asthma attack during a school event. The student is wheezing and clutching their chest. Should the priority be restoring airflow or calming the heart rate? 3. Making Generalization <i>At this stage of the lesson, the teacher guides students to consolidate their understanding of how different body systems interact to maintain function and respond to stress or injury</i> Option 1: Connecting the Dots <i>The teacher facilitates a reflective activity using sticky notes. Students are asked to write down three key takeaways from the lesson—focusing on how body systems interact, respond to injury, and compensate during conditions like a torn ACL or an asthma attack.</i> <i>Each student posts their notes on the board. As a class, they then group similar ideas to identify patterns and shared insights. With guidance from the teacher, students work together to craft a summary statement that captures the essential principle of the lesson:</i> Option 2: One Big Idea-System Coordination <i>Each student is asked to distill everything they’ve learned into a single generalization statement that answers the question: “What is the most important thing I learned about how damage to one part of a body system affects the whole organism?” The statements shared aloud, posted on the board, or collected as exit tickets to assess conceptual mastery.</i> 4. Evaluating Learning <i>Now that we have explored, analyzed, and applied our learning, it is time to assess how well we have understood and integrated these concepts.</i>	<i>* Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. It supports varied learning styles.</i> <i>Teachers may select one or more of the suggested options to implement differentiated instruction based on their students' needs, learning styles, and classroom context. It supports varied learning styles.</i>
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Option 1: Modified TRUE or FALSE

The teacher will instruct the students to write “TRUE” if the statement is correct. If the statement is incorrect, write “FALSE” and modify the statement to make it accurate.

1. The skeletal system only provides structure and does not interact with other systems.
2. Muscles can function properly without the support of bones.
3. A torn ACL affects only the skeletal system and has no impact on other systems.
4. During an asthma attack, the respiratory system is the only system affected.
5. System coordination allows the body to respond to damage and maintain function.

Option 2: Link and Think

Students match each item in Column A with the correct description in Column B. After matching, they will choose two pairs and write a short explanation of how they show system coordination or damage impact.

Column A (Condition/ Structure)	Column B (System Impact or Function)
1. Torn ACL	A. Muscles compensate for joint instability
2. Asthma Attack	B. Reduced oxygen forces heart to work harder
3. Skeletal System	C. Provides structure and attachment for movement
4. Respiratory System	D. Delivers oxygen and removes carbon dioxide
5. System Coordination	E. Body systems interact to maintain overall function

5. Additional Activities

To reinforce conceptual understanding and allow for creative expression, the teacher asks the students to do an individual performance task.

	<table><tr><td>Accuracy of Information</td><td>All facts are accurate and well-supported by examples or evidence.</td><td>Most facts are accurate; minor errors present.</td><td>Many inaccuracies; lacks factual support.</td></tr><tr><td>Visual or Written presentation</td><td>(Infographic) Visually engaging, well-labeled, and informative. (Essay) Well-written, grammatically correct, and within word count.</td><td>(Infographic) Clear visuals with minor design flaws. (Essay) Mostly well-written with few errors.</td><td>(Infographic) Poor visuals or missing elements. (Essay) Writing is unclear or incomplete.</td></tr><tr><td>Critical Thinking and Insight</td><td>Offers thoughtful analysis of how systems support recovery and adapt to stress.</td><td>Shows some insight into system responses and recovery.</td><td>Lacks analysis; superficial or off-topic.</td></tr></table>	Accuracy of Information	All facts are accurate and well-supported by examples or evidence.	Most facts are accurate; minor errors present.	Many inaccuracies; lacks factual support.	Visual or Written presentation	(Infographic) Visually engaging, well-labeled, and informative. (Essay) Well-written, grammatically correct, and within word count.	(Infographic) Clear visuals with minor design flaws. (Essay) Mostly well-written with few errors.	(Infographic) Poor visuals or missing elements. (Essay) Writing is unclear or incomplete.	Critical Thinking and Insight	Offers thoughtful analysis of how systems support recovery and adapt to stress.	Shows some insight into system responses and recovery.	Lacks analysis; superficial or off-topic.	<i>suit their learning objectives and student needs.</i>
Accuracy of Information	All facts are accurate and well-supported by examples or evidence.	Most facts are accurate; minor errors present.	Many inaccuracies; lacks factual support.											
Visual or Written presentation	(Infographic) Visually engaging, well-labeled, and informative. (Essay) Well-written, grammatically correct, and within word count.	(Infographic) Clear visuals with minor design flaws. (Essay) Mostly well-written with few errors.	(Infographic) Poor visuals or missing elements. (Essay) Writing is unclear or incomplete.											
Critical Thinking and Insight	Offers thoughtful analysis of how systems support recovery and adapt to stress.	Shows some insight into system responses and recovery.	Lacks analysis; superficial or off-topic.											
V. ASSESSMENT <i>(Assessing Learnings)</i>	<i>The teacher instructed the students to read each question carefully and select the best answer based on what they have learned.</i> <i>Directions. Read each item carefully and write the letter of the correct answer in a separate sheet of paper</i> <i>1. Which body systems are directly affected by a torn ACL injury?</i> <i>A. Respiratory and circulatory systems </i>													

	C. Systems ignore damage and continue functioning normally D. Each system works independently without affecting others 6. How does the body compensate when one system is damaged? A. Other systems adjust and support the damaged system to maintain function B. The body stops functioning entirely C. The damaged system shuts down completely D. Other systems ignore the damage 7. Why does the muscular system become involved when the skeletal system is injured, such as in a torn ACL? A. Because muscles are part of the digestive system B. Because muscles produce hormones to heal bones C. Because muscles help stabilize and support movement when joints are damaged D. Because muscles control breathing and heart rate 8. How does the circulatory system respond during an asthma attack? A. It slows down to conserve energy B. It increases heart rate to deliver limited oxygen more efficiently C. It produces mucus to clear airways D. It stops pumping blood to the 9. Which scenario best demonstrates effective system coordination in response to injury? A. A person with a torn ACL continues walking without any muscular adjustment B. The digestive system increases activity during a respiratory emergency C. During an asthma attack, the respiratory system fails while other systems remain unaffected D. After a torn ACL, muscular system adjusts movement to stabilize the knee while the nervous system signals pain 10. What does ACL stand for in the human body? A. Abdominal Cartilage Layer B. Automatic Control Ligament C. Anterior Cruciate Ligament D. Arterial Circulation Line
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>

APPENDIX

Modified True or False Quiz

1. **False** → *The skeletal system not only provides structure but also interacts with other systems, such as the muscular system for movement and the circulatory system through bone marrow production of blood cells.*
2. **False** → *Muscles rely on bones for attachment and leverage to produce movement. Without the skeletal system, voluntary muscle movement would not be possible.*
3. **False** → *A torn ACL affects both the skeletal and muscular systems. The ligament is part of the skeletal system, but the injury also impacts muscles that must compensate for joint instability.*
4. **False** → *While the respiratory system is primarily affected during an asthma attack, the circulatory system is also impacted due to reduced oxygen levels, causing the heart to work harder.*
5. **True** → *System coordination refers to the way body systems work together to adapt and maintain stability when one part is damaged*

Assessment

1. *D. Skeletal and muscular systems Explanation: The ACL (anterior cruciate ligament) is part of the knee joint, which belongs to the skeletal system. When torn, it destabilizes the joint, and the muscular system must compensate to support movement and prevent further injury.*
2. *B. Respiratory and circulatory systems Explanation: The respiratory system struggles to bring in oxygen due to airway constriction. The circulatory system responds by increasing heart rate and blood flow to deliver the limited oxygen to vital organs.*
3. *D. It connects bones, muscles, affecting movement. Explanation: The ACL stabilizes the knee by connecting bones. When damaged, it disrupts skeletal integrity and forces muscles to adjust movement patterns to maintain balance and prevent further harm.*
4. *D. The respiratory system struggles to bring in oxygen, and the circulatory system works harder to deliver it. Explanation: Asthma narrows airways, reducing oxygen intake. The heart compensates by pumping faster to circulate the available oxygen efficiently, showing coordinated system response.*
5. *B. Multiple systems interact and adjust to maintain body function Explanation: System coordination refers to how different body systems communicate and support each other during stress or injury. For example, when one system is impaired, others adapt to preserve overall function and promote recovery.*
6. *A. Other systems adjust and support the damaged system to maintain function Explanation: The body adapts by activating other systems to stabilize function and promote recovery.*

7. *C. Because muscles help stabilize and support movement when joints are damaged Explanation: Muscles compensate for joint instability by adjusting movement and providing support.*
8. *B. It increases heart rate to deliver limited oxygen more efficiently Explanation: To compensate for reduced oxygen intake, the heart pumps faster to ensure delivery to vital organs.*
9. *D. After a torn ACL, muscular system adjusts movement to stabilize the knee while the nervous system signals pain Explanation: This shows multiple systems working together—muscles stabilize, nerves alert the body to injury.*
10. *C. Anterior Cruciate Ligament Explanation: The ACL is a key ligament in the knee that connects the thighbone to the shinbone and stabilizes movement.*